This project focuses on exploring and also simulating communication protocols used in IIoT. The main goal of this project was to understand how the sensors are able to communicate and give data using protocols.

I worked with simulation scripts that demonstrate how the sensors send and receive data. We also analyzed the outputs that the simulations gave us and how it handles communication between devices.

Some learning outcomes I learned is,

MQTT is lightweight protocol that uses a published model. Using a central broker, devices can efficiently exchange data. This is very efficient for IoT systems with many devices communicating together at the same time

CoAP is designed for constrained devices with low resources. An example can be battery powered sensors. This allows sensors to send data with minimal productive cost.

OPC UA is commonly used a lot in industrial systems. They provide more secure communication and data exchange. OPC focus more on reliability and better industrial communications.

Some challenges included how different protocols operate and how the scripts represent real sensor communication.

To overcome this challenge, We had to review the python scrips and break down the code step by step.

The knowledge I gained from this project can be applied for future AI-Based projects that involve device communication or even sensor collection for data. Some improvements or next step on this project is by analzying more on performance metrics like latency to gain more details